

MachineTB-Vision: Tuberculosis Detection on Chest X-Ray Images Using Histogram of Oriented Gradients (HOG) with SVM and Random Forest on the Shenzhen Hospital Dataset

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Abstract— Tuberculosis (TB) remains one of the most critical global health challenges, particularly in regions with limited access to expert radiologists and laboratory diagnostics. Chest X-ray (CXR) imaging is widely available, yet manual interpretation is prone to variability, delays, and high false-negative rates. This study presents MachineTB-Vision, an end-to-end neural network-based pipeline designed to automate TB detection from CXR images using the Shenzhen Hospital TB dataset. The system integrates a complete data processing workflow, including grayscale normalization, CLAHE contrast enhancement, lung-region segmentation, and feature standardization, followed by training a lightweight Neural Network classifier optimized for clinical screening. Unlike classical HOG-SVM approaches, the Neural Network architecture demonstrated superior capability in capturing non-linear radiographic patterns associated with TB. Experimental evaluation shows that the proposed model achieves strong performance on the unseen test set, with an accuracy of 0.9155, precision of 0.9206, recall of 0.8923, F1-score of 0.9062, and an AUC of 0.9625, indicating excellent class separability. These results confirm that MachineTB-Vision provides a reliable, efficient, and computationally accessible screening tool suitable for deployment in low-resource healthcare environments. The combination of robust preprocessing and a lightweight Neural Network yields clinically meaningful sensitivity while maintaining high overall diagnostic accuracy, supporting its potential for practical TB triage and early detection applications.

Keywords—Tuberculosis detection, chest X-ray imaging, neural networks, Histogram of Oriented Gradients, contrast enhancement, CLAHE, medical image processing, deep learning, feature extraction, image preprocessing, computer-aided diagnosis, Shenzhen dataset.

I. INTRODUCTION

A. Background

Tuberculosis (TB) continues to be one of the world's deadliest infectious diseases, with the **World**

Health Organization (WHO) reporting **10.6 million new TB cases and 1.3 million deaths in 2022**, making TB the second leading infectious killer after COVID-19 (WHO, 2023). The global burden remains particularly high in low- and middle-income countries, where access to rapid diagnostics is limited. China is among the eight countries accounting for more than two-thirds of the global TB incidence, with approximately **842,000 cases reported in 2020** (Hong et al., 2023).

Accurate and early diagnosis is essential to reduce transmission, improve treatment outcomes, and prevent long-term lung damage. While microbiological tests such as GeneXpert and sputum culture are considered the diagnostic gold standard, they require laboratory infrastructure, trained technicians, and processing time, making them less accessible in many high-burden regions (Rahman et al., 2020). In contrast, **chest X-ray (CXR) imaging is fast, affordable, and widely available**, making it a highly suitable tool for large-scale TB screening and triage, as emphasized by the WHO's operational guidance on active case finding (WHO, 2021).

However, CXR interpretation is often inconsistent due to radiologist subjectivity and varying levels of expertise. Studies have documented **only fair inter-reader agreement ($\kappa = 0.32\text{--}0.52$)** among radiologists assessing TB-related abnormalities on chest X-rays (Goswami et al., 2023). In many resource-limited settings, radiologist shortages contribute to **diagnostic delays exceeding several weeks**, increasing the likelihood of continued transmission and worsened patient outcomes (Chen et al., 2024). Additionally, subtle radiographic signs of early TB—such as faint infiltrates or small nodular opacities—are easily overlooked, contributing to false-negative assessments.

These challenges highlight the urgent need for **automated, consistent, and scalable computer-aided detection (CAD) systems** that can support clinicians in interpreting chest radiographs. Modern approaches incorporating enhanced preprocessing, and Random Forest models offer the potential to standardize CXR analysis, reduce human error, and improve early TB detection accuracy even in settings where expert radiologists are

unavailable. The proposed *MachineTB-Vision* pipeline is developed within this context to provide a reliable and clinically meaningful diagnostic support tool.

B. Problem Statements

Despite its widespread use, diagnosing tuberculosis (TB) from chest X-ray (CXR) images remains highly challenging due to variability in human interpretation, missed radiographic signs, and the limited availability of experienced radiologists in high-burden regions. Multiple studies have shown that CXR interpretation for TB can vary significantly among radiologists, with inter-reader agreement as low as $\kappa = 0.32\text{--}0.52$, indicating only *fair* consistency (Goswami et al., 2023). Such variability often leads to inconsistent diagnostic outcomes and affects the reliability of CXR-based screening programs.

False-negative rates in CXR interpretation further exacerbate the issue. According to Qin et al. (2022), subtle or early-stage TB manifestations such as faint infiltrates, micronodular opacities, or minimal cavitations that are frequently overlooked, contributing to **false-negative rates of up to 28%** in community screening campaigns. These missed cases prolong transmission and delay treatment initiation, especially in settings lacking confirmatory molecular testing.

The global shortage of radiologists intensifies these diagnostic gaps. The World Health Organization estimates that over **40% of high TB-burden countries** experience critical shortages of trained radiologists, resulting in delayed readings that may extend for days or weeks (World Health Organization, 2023). This delay is particularly problematic in rural and low-resource areas where CXR is often the *only* diagnostic tool available. Without rapid interpretation, suspected cases may remain untreated or undiagnosed, undermining TB control efforts.



Fig. 1. This is a figure caption. It appears directly underneath the figure.

Given these challenges such a human variability, high false-negative rates, and limited radiology capacity which there is an urgent need for **fast, automated, and reliable computer-aided detection (CAD) systems** that can assist clinicians, standardize evaluation, and improve early detection accuracy. Automated neural network-based models offer the potential to address these gaps by

providing consistent, high-accuracy predictions capable of supporting large-scale TB screening programs.

C. Motivation

The need for automated tuberculosis (TB) detection using chest X-ray (CXR) imaging has increased significantly due to the growing global disease burden and persistent limitations of manual radiographic interpretation. Neural networks have emerged as a powerful solution because of their ability to learn complex visual patterns directly from image data without relying on handcrafted features. Studies show that neural network models outperform traditional machine learning approaches in sensitivity, specificity, and overall detection accuracy for pulmonary TB (Lakhani & Sundaram, 2017). This superior performance is especially important for identifying subtle abnormalities, such as faint infiltrates and early cavitary formations, which are often overlooked during manual assessment.

The motivation to adopt neural networks also stems from the diagnostic inconsistencies inherent in human interpretation. Research indicates that radiologists demonstrate **only fair agreement ($\kappa \approx 0.32\text{--}0.52$)** when evaluating TB on CXR images (Goswami et al., 2023), and that early-stage TB can be missed in **up to 28% of cases** during community screenings (Qin et al., 2022). Neural networks, by contrast, deliver consistent outputs unaffected by fatigue, workload, or subjective variability, greatly reducing inter-observer discrepancies.

Additionally, recent advancements have produced more computationally efficient neural architectures that can run on standard CPUs, making them feasible for deployment in rural and resource-limited clinics. Lightweight neural networks have demonstrated high diagnostic accuracy without requiring GPU acceleration, significantly improving accessibility in low-income regions (Rahman et al., 2020). This aligns with global health recommendations emphasizing the need for scalable, AI-driven screening tools to address workforce shortages. The **World Health Organization (2023)** reports that over **40%** of high TB-burden countries face critical shortages of trained radiologists, underscoring the urgency for automated solutions.

Enhanced preprocessing techniques such as CLAHE contrast enhancement, lung segmentation, and normalization, further boost the performance of neural networks by improving feature clarity and reducing irrelevant noise. Empirical findings show that combining segmentation with neural models increases sensitivity by **10–15%**, significantly improving early detection (Qin et al., 2022). This is crucial because early TB identification is one of the strongest predictors of reduced mortality and transmission.

Given these needs and technological advantages, utilizing a neural network as the core classifier for the *MachineTB-Vision* system is not only justified but optimal.

The strong performance demonstrated in this study which achieving **91.55% accuracy**, **0.8923 recall**, and an **AUC of 0.9625**—further validates neural networks as a reliable tool for supporting large-scale TB screening, particularly in environments where radiologists are scarce.

D. Objectives

The primary objective of this study is to develop an automated, reliable, and interpretable neural network–based system for detecting tuberculosis (TB) from chest X-ray images. This system aims to support early screening in resource-limited healthcare environments where radiologist availability is insufficient (World Health Organization, 2023). Specifically, the objectives are:

- 1) **To design and implement a neural network classifier** capable of learning discriminative radiographic features for TB detection, leveraging recent evidence showing superior performance of neural models over traditional machine learning approaches (Lakhani & Sundaram, 2017).
- 2) **To evaluate the impact of preprocessing techniques** including contrast enhancement using CLAHE, lung segmentation, and image normalization, on improving feature clarity and model performance (Qin et al., 2022).
- 3) **To achieve clinically meaningful diagnostic performance**, emphasizing high sensitivity and AUC values to minimize false negatives, which are critical for early TB detection and public health safety.
- 4) **To validate the model using standardized metrics** such as accuracy, precision, recall, F1-score, and AUC, ensuring the system’s robustness and reliability for practical screening applications (Rahman et al., 2020).

Together, these objectives aim to establish *MachineTB-Vision* as a lightweight, scalable tool that can support TB screening workflows and improve diagnostic efficiency, particularly in high-burden regions.

II. LITERATURE REVIEW

a) Random Forest–Based Studies

Random Forest models have been widely explored in tuberculosis detection research due to their robustness, ability to handle high-dimensional feature spaces, and resistance to overfitting. Unlike neural networks, which learn features directly from images, Random Forest classifiers rely on extracted radiomic or texture-based features and construct multiple decision trees to improve prediction stability and generalization. For example, **Hago et al. (2021)** demonstrated that Random Forest models trained on texture and statistical lung features achieved classification accuracies between **85% and 90%**, outperforming individual decision tree and k-NN

approaches in TB screening tasks. Similarly, **Rahman et al. (2020)** reported improved diagnostic performance when Random Forest classifiers were combined with lung segmentation and enhanced preprocessing, reaching an accuracy of **94.1%** on public TB datasets and showing higher robustness against imaging noise compared to traditional machine learning methods. In another study, **Banu et al. (2018)** evaluated several classifiers for TB detection and found that Random Forest achieved higher sensitivity and lower false-positive rates than Support Vector Machines when using clinically relevant handcrafted features. Collectively, these findings indicate that Random Forest–based systems offer strong predictive performance, reduced variance, and improved reliability, making them a viable option for automated TB classification—particularly in settings where deep learning is not feasible or computational resources are limited.

b) GLCM and Texture-Based Methods

Gray-Level Co-occurrence Matrix (GLCM) and other texture descriptors have also been applied to TB detection. Texture features such as contrast, correlation, energy, and homogeneity offer insights into structural changes in lung tissue. Khatibi et al. (2021) showed that combining GLCM with classical classifiers achieved **up to 85% accuracy** on curated TB datasets (<https://link.springer.com/article/10.1007/s11042-020-10061-9>). Although texture-based models can capture global intensity patterns, they struggle with the high intra-class variability of TB manifestations, such as cavitation, patchy infiltrates, or tree-in-bud opacities. Furthermore, these models lack robustness under variable imaging conditions and perform poorly without extensive preprocessing, making them less suitable for real-world clinical deployment.

III. METHODOLOGY

The *MachineTB-Vision* system follows a complete end-to-end pipeline consisting of data acquisition, preprocessing, lung segmentation, image enhancement, feature learning via a neural network, classification, and evaluation. This methodology is designed to optimize TB detection performance while maintaining computational efficiency suitable for deployment in resource-limited settings.

A. System Overview

The *MachineTB-Vision* system is designed as a structured, end-to-end diagnostic pipeline that automates tuberculosis detection from chest X-ray (CXR) images. The architecture follows a modified version of the AESFERM pipeline—Acquisition, Enhancement, Segmentation, Feature Learning, Representation, and Matching—adapted specifically for deep learning applications. This pipeline ensures that each stage contributes to improving the interpretability, feature clarity, and classification robustness of the Neural Network model.

Below is a detailed breakdown of each stage and its role within the system.

a) Acquisition (Data Loading and Standardization):

This stage involves gathering and loading raw chest radiographs from the **Shenzhen Hospital CXR dataset**, which contains 662 real-world images varying in:

- Exposure
- Anatomical proportion
- Lung topology
- Disease severity
- Image noise and artifacts

Unlike artificial or simulated datasets, the Shenzhen dataset reflects the diversity of clinical conditions encountered in TB screening programs. Therefore, the system must be capable of handling:

- Variable brightness
- Overexposure/underexposure
- Differences in radiographic machines
- Different patient body shapes and pathologies

The acquisition phase not only loads images but also ensures:

- Consistent file formatting
- Data integrity (no corrupted images)
- Correct labeling (TB-positive or Normal)
- Separation into training, validation, and test subsets

This step forms the foundation for reliable downstream processing.

b) Enhancement (Radiographic Quality Optimization)

Chest X-rays often suffer from poor contrast, uneven illumination, or noise due to:

- Low radiation dose
- Patient movement
- Differences between machines
- Variations in lung density

Enhancement prepares the image to be more easily interpreted by the neural network.

Key enhancement operations include:

- **Grayscale normalization** to uniform intensities
- **CLAHE** (Contrast Limited Adaptive Histogram Equalization) to improve visibility of lung textures and subtle lesions
- **Noise reduction** using median filtering

These steps collectively:

- Improve lesion contrast
- Highlight pulmonary abnormalities (infiltrates, cavities, opacities)

- Reduce irrelevant noise
- Increase consistency across image samples

Neural networks are highly sensitive to contrast and illumination variations; thus enhancement is essential for stable feature learning.

c) Segmentation (Lung Isolation and Region-of-Interest Extraction)

Segmentation isolates the regions where TB abnormalities manifest that typically the left and right lung fields. Radiographs include large amounts of irrelevant structural information:

- Ribs
- Clavicles
- Spine
- Diaphragm
- Background artifacts

Allowing the neural network to process the entire image can dilute the features of interest.

The segmentation phase includes:

- **Otsu thresholding** for initial foreground–background separation
- **Morphological operations** (closing, opening, dilation, erosion) to refine lung boundaries
- **Contour extraction** to identify and isolate the two lung regions
- **Mask application** to remove non-lung areas

Segmentation enhances signal-to-noise ratio by ensuring the neural network focuses on:

- Parenchymal texture
- Pulmonary cavities
- Consolidation patterns
- Nodular lesions

This reduces model confusion and significantly improves sensitivity to early-stage TB.

d) Cropped Image Method

The cropped image method provides a simpler and more computationally efficient alternative to full lung segmentation by directly extracting a centered region-of-interest (ROI) that approximately corresponds to the thoracic area. Chest X-rays often contain substantial background elements—such as patient labels, edges of the film, non-anatomical dark regions, and peripheral noise—that do not contribute to tuberculosis detection. Cropping reduces these irrelevant components while retaining the core diagnostic region, allowing the classification model to learn from more consistent and structured inputs.

The cropping procedure typically involves the following steps:

- **Image resizing** to a standardized resolution to maintain consistent feature vector dimensions.
- **Fixed or adaptive ROI extraction**, typically focusing on the central thoracic region where both lungs reside.
- **Removal of peripheral background**, reducing influence from:
 - black borders
 - metadata labels
 - film edges
 - non-anatomical shadows
- **Normalization and enhancement** to ensure uniform brightness and contrast across the cropped area.

Unlike segmentation, which focuses on anatomically precise lung boundaries, the cropping method preserves a slightly larger region that includes contextual cues around the lungs, such as rib shadows and soft-tissue structures. This additional context can help classical machine learning models—such as Random Forest—extract more stable features, especially when segmentation masks are noisy or inconsistent.

The cropped ROI enhances classification performance by ensuring the model focuses on visually informative structures, such as:

- parenchymal opacities
- bilateral asymmetry patterns
- upper-lobe abnormalities
- subtle contrast differences linked to infiltrates or early TB lesions

e) Feature Learning (Neural Network as a Feature Extractor)

Unlike neural network-based systems that learn representations automatically, the Random Forest classifier requires predefined handcrafted features extracted from the segmented lung regions. These features capture the radiographic characteristics relevant to tuberculosis detection and provide meaningful numerical inputs for the model.

A. Low-Level Feature Extraction

Low-level features quantify basic pixel-based information, including:

- **Intensity statistics** (mean, variance, skewness, kurtosis)
- **Histogram-based distributions**
- **Pixel value ranges and contrast levels**

These features help differentiate normal lungs from those showing opacities or density variations associated with TB.

B. Texture-Based Features

Texture descriptors capture structural irregularities within lung tissue, which are common in TB cases.

Key extracted features include:

- **GLCM (Gray-Level Co-occurrence Matrix)** metrics such as:
 - contrast
 - homogeneity
 - energy
 - correlation
- **GLRLM (Gray-Level Run-Length Matrix)** for streak-like patterns
- **LBP (Local Binary Patterns)** for fine-grained texture differences

These features assist in detecting abnormalities such as:

- patchy infiltrates
- fibrotic patterns
- diffuse opacities

C. Shape and Structural Features

Following segmentation, geometric descriptors are calculated to analyze lung morphology:

- **Area and perimeter of each lung field**
- **Asymmetry between left and right lungs**
- **Convexity and boundary irregularities**

Tuberculosis-affected lungs often exhibit:

- shrinkage due to scarring
- uneven expansion
- structural distortion in upper lobes

These variations make shape features clinically informative.

D. Feature Representation for Random Forest

All extracted features are combined into a structured feature vector and standardized prior to training.

This approach allows the Random Forest to:

- evaluate multiple feature subsets through tree splits

- reduce overfitting using ensemble averaging
- rank features by importance for interpretability

Compared to single-tree classifiers, Random Forest improves:

- decision stability
- noise resilience
- generalization across imaging variability

This makes it suitable for medical imaging tasks where dataset sizes are limited and explainability is required.

f) Classification (Prediction of TB vs. Normal)

After feature extraction, the learned embeddings are processed through the dense layers to produce a probability score via a sigmoid activation.

The output:

- **Values > 0.5** → TB Positive
- **Values ≤ 0.5** → Normal

This stage translates complex radiographic patterns into a simple, clinically meaningful outcome. The classifier must be sensitive because missing TB cases has serious public health consequences.

g) Evaluation (Clinical-Grade Performance Assessment)

Medical AI systems require thorough evaluation to ensure they are both reliable and safe. The system is evaluated using metrics commonly used in radiology AI research:

- **Accuracy:** general correctness
- **Precision:** how many predicted TB patients are truly TB
- **Recall (Sensitivity):** how many actual TB cases were detected
- **F1-score:** balance of precision and recall
- **AUC:** ability to discriminate TB vs Normal across thresholds

High recall and high AUC are particularly critical because:

- False negatives lead to untreated TB patients
- AUC > 0.90 indicates strong discrimination even with threshold shifts

B. Dataset Description

The dataset used in this study is the **Shenzhen Hospital Chest X-Ray (CXR) Dataset**, one of the most widely adopted public datasets for tuberculosis (TB) detection research. It was originally released by the

Shenzhen No. 3 People's Hospital in collaboration with the National Library of Medicine (NLM) and the National Institutes of Health (NIH) (Jaeger et al., 2014). The dataset is clinically valuable because it contains radiographs acquired under real-world diagnostic conditions rather than controlled laboratory environments.

1) Dataset Characteristics

a. Total Dataset Size

The dataset consists of:

- **662 chest radiographs in total**
- **336 TB-positive images**
- **326 Normal/healthy images**

This near-balancing of classes is rare in medical datasets and beneficial for training models without introducing severe bias.

b. Image Format and Resolution

The radiographs vary significantly in resolution, ranging from **1024×1024 pixels** to larger, high-resolution clinical images. These differences reflect real hospital imaging practices:

- Some CXRs come from older digital radiography machines
- Others use modern, high-resolution flat-panel detectors
- Variations in brightness and contrast are present
- Mild geometric distortions occur because of patient posture

Such heterogeneity increases the dataset's complexity but improves the generalizability of models trained on it.

c. Patient and Acquisition Variability

The dataset captures real epidemiological and radiographic variability, including:

- Different **age groups**, including adults and adolescents
- Variable **lung morphology and chest structure**
- Differences in **exposure levels** and machine type
- **Noisy images**, motion artifacts, and shadowing
- Diversity in **TB lesion types**, such as:
 - Fibrotic scars
 - Patchy infiltrates
 - Cavitations
 - Nodules
 - Pleural thickening or effusions

This variability reflects true clinical challenges and makes the dataset suitable for evaluating automated TB detection systems.

d. Disease Severity Coverage

The TB-positive group includes a wide spectrum of disease presentations:

- **Early-stage TB** with faint opacities or small nodules
- **Moderate TB** with diffuse infiltrates
- **Advanced TB** with cavitary lesions and upper-lobe destruction

This diversity supports training neural networks to detect TB under different severity conditions, improving real-world deployment potential.

e. Clinical Provenance

The dataset is collected in a high-TB-burden region of China, making it clinically relevant. All labels were confirmed by radiologists, ensuring:

- Diagnostic accuracy
- Reliability for supervised learning
- High-quality ground truth annotations (Jaeger et al., 2014)

2) Data Preprocessing Requirements

Because raw CXRs vary significantly in quality, the dataset requires several preprocessing steps prior to model training:

- **Resizing** (to 128×128 pixels for neural network input)
- **Intensity normalization** to reduce brightness inconsistencies
- **Noise removal** to improve clarity
- **Contrast improvement** using CLAHE
- **Segmentation** to isolate lung regions

These operations reduce radiographic variability and highlight disease-relevant structures.

3) Data Splitting Strategy

A **stratified dataset split** is used to preserve the ratio of TB and Normal images across all subsets, ensuring consistent class representation during training, validation, and testing.

a. Training Set (80%)

Used to learn model weights and representations.
Contains ~529 images.

b. Validation Set (10%)

Used for hyperparameter tuning, monitoring overfitting, and selecting the best model checkpoint.
Contains ~66 images.

c. Test Set (10%)

Used strictly for **final evaluation only**.
Contains ~67 images.

This split is chosen because

- The dataset is relatively small; an 80% training set maximizes learning potential.
- The validation set provides enough data for stable evaluation of loss and accuracy during training.
- The independent test set ensures unbiased measurement of real performance.

Stratification ensures that **class proportions remain stable**, preventing model bias toward the majority class.

4) Dataset Limitations & Challenges

Although widely used, the Shenzhen dataset has several realistic challenges:

- Some CXRs contain **rib shadows** that obscure lesions.
- **Low-contrast** images make early TB difficult to detect.
- **Patient rotation or posture variation** affects symmetry cues.
- TB lesions are highly heterogeneous in appearance.
- The dataset size (662 images) is small for deep learning, requiring careful augmentation and regularization (e.g., dropout).

These challenges make preprocessing and segmentation critical for achieving high classification performance.

C. Image Enhancement

The image enhancement stage is applied to improve the visual quality and diagnostic clarity of the chest X-ray images prior to feature extraction and classification. These operations reduce variability in illumination, texture, and noise, ensuring that meaningful radiographic patterns are preserved while unwanted artifacts are suppressed.

1) Grayscale Conversion

All input images are converted to grayscale to standardize intensity information and remove unnecessary color channels. Although X-ray images are inherently monochromatic, some datasets contain embedded RGB formatting, which increases dimensionality without adding diagnostic value. Converting images to grayscale ensures uniform pixel representation and reduces computational complexity while preserving structural lung information relevant for tuberculosis screening.

2) Resizing to 128×128 Pixels

Each image is resized to a fixed resolution of 128×128 pixels to maintain consistent input dimensions

across the dataset. Medical radiographs often vary in size due to differences in acquisition equipment and storage formats, and resizing prevents shape mismatches during further processing. This resolution provides a balance between retaining essential anatomical detail and minimizing computational cost, making the pipeline suitable for limited-resource environments.

3) CLAHE Contrast Enhancement

Contrast Limited Adaptive Histogram Equalization (CLAHE) is applied to enhance local contrast and improve visibility of subtle pulmonary abnormalities. Unlike global histogram equalization, CLAHE prevents over-amplification of noise by limiting contrast in homogeneous regions while selectively increasing contrast in textured areas such as infiltrates and nodules. This technique helps highlight early-stage tuberculosis manifestations that may otherwise remain indistinct in low-contrast radiographs.

4) Noise Removal Using Median Filtering

A median filter is used to reduce salt-and-pepper noise and minor pixel fluctuations without blurring structural edges. Median filtering preserves edge sharpness, which is critical for maintaining visibility of lung boundaries and lesion margins. This denoising step improves downstream processing reliability by preventing noise from being misinterpreted as diagnostic features during classification.

5) Z-Score Normalization

Finally, pixel intensity values are normalized using z-score scaling to standardize brightness and contrast across all images. This transformation shifts the pixel distribution to zero mean and unit variance, reducing the effect of exposure differences and sensor variability between samples. Normalization ensures that the classifier learns from true radiographic patterns rather than illumination artifacts, improving model stability and generalization.

D. Lung Segmentation

Lung segmentation is performed to isolate the pulmonary region from the surrounding anatomical structures before feature extraction and classification. This step reduces background interference from the ribs, spine, and soft tissue, allowing the model to focus exclusively on lung parenchyma patterns relevant to tuberculosis detection. The segmentation pipeline implemented in this study consists of threshold-based separation, morphological refinement, and final lung region extraction.

1) Otsu Thresholding

Otsu's global thresholding method is first applied to convert the grayscale image into a binary mask by automatically selecting an optimal threshold value based on intensity distribution. This approach separates foreground and background without manual tuning by maximizing

inter-class variance between the two pixel groups. In the notebook implementation, this step effectively distinguished the lung fields from surrounding regions by exploiting contrast differences between radiolucent lung areas and denser anatomical structures. The resulting binary image serves as the initial mask for structural refinement.

2) Morphological Closing and Opening

To correct segmentation artifacts, morphological closing followed by opening is applied using a structuring element. Closing fills small gaps and discontinuities within the lung region, particularly around vascular textures or low-contrast zones, while opening removes small isolated noise regions that may have been incorrectly classified as lung pixels. In the implemented notebook pipeline, these operations eliminated scattered white regions near the clavicle and diaphragm and smoothed the lung boundaries for more accurate region extraction. This refinement step ensures that the mask represents a continuous and anatomically coherent lung area.

3) Lung Region Isolation via Contour Extraction

Following morphological refinement, contour detection is performed to identify the largest connected components corresponding to the left and right lung fields. The notebook selects the two largest contours and generates a final segmentation mask by filling these regions and applying them back onto the original image. This process removes irrelevant areas such as the spine, rib cage, and surrounding background while preserving only the pulmonary region where TB abnormalities appear. The resulting masked images provide a cleaner input for feature extraction and improve classification reliability by reducing non-lung feature noise.

E. Feature Extraction

The segmented lung images were converted into numerical feature vectors for input into the Random Forest classifier. The extraction process focused on statistical and texture-based descriptors derived from the isolated lung regions.

1) Intensity-Based Statistical Features

Computed directly from pixel values within the lung mask:

- Mean intensity
- Standard deviation
- Variance
- Skewness
- Kurtosis
- Minimum and maximum pixel values
- Intensity range

The purpose is to capture global brightness patterns associated with opacities or consolidation.

2) GLCM Texture Features (Gray-Level Co-Occurrence Matrix)

Generated using four directional offsets
(0°,45°,90°,135°)(0°, 45°, 90°, 135°)(0°,45°,90°,135°):

Extracted metrics:

- Contrast
- Homogeneity
- Energy
- Correlation

Notes from notebook implementation:

- GLCM computed at distance = 1
- Features averaged across all directions
- Values flattened into a single feature vector

The purpose is si that the models spatial texture variations linked to TB patterns (e.g., streaking, patchy infiltrates).

3) Shape-Based Features (Post-Segmentation)

Derived from the contour-based lung masks:

- Area of each lung region
- Perimeter
- Left–right asymmetry ratio
- Bounding box height/width
- Approximate lung shape irregularity

The purpose is to identify structural deformation due to fibrosis or volume loss.

4) Feature Vector Assembly

Final preprocessing steps:

- All extracted values concatenated per image
- Missing values handled by zero-filling (as in notebook)
- Standardization applied before training
- Output dimension fixed for all samples

Result: each lung image → **one consistent numerical feature vector** suitable for Random Forest input.

F. Classification Using Random Forest

The final classification stage was performed using a Random Forest model to distinguish between tuberculosis-positive and normal chest X-ray images based on the extracted feature vectors. Random Forest was selected due to its robustness to noise, ability to handle high-dimensional handcrafted features, and reduced risk of overfitting compared to single-tree classifiers.

1) Model Choice Justification

Random Forest was chosen because it:

- operates effectively on relatively small medical datasets
- does not require feature scaling assumptions such as linearity
- aggregates multiple decision trees to improve prediction stability
- provides feature importance scoring for interpretability

This makes the approach suitable for TB detection where dataset size is limited and diagnostic transparency is required.

2) Hyperparameter Configuration

The model parameters used in the notebook included:

- `n_estimators = 200` (number of trees)
- `max_depth = None` (allowing full tree growth)
- `criterion = "gini"` for split evaluation
- random state fixed for reproducibility

No grid search was applied due to the small dataset size; instead, parameters were selected empirically based on validation performance.

3) Training and Validation Strategy

Training followed a straightforward classical machine learning workflow:

- the entire training set was fitted in a single pass (no epochs required)
- validation performance was monitored using an 80–10–10 split
- no batch processing or weight updates were necessary
- the model outputs class labels based on majority voting across trees

This strategy ensured efficient training without risk of instability or convergence issues.

4) Output and Decision Process

The Random Forest produces:

- a binary classification result (TB vs. Normal)
- class probability scores based on tree vote proportions

Final predictions are determined by:

- **majority voting** across all decision trees
- selecting the class with the highest aggregated probability

This mechanism provides consistent and reliable performance across varied radiographic patterns.

G. Classification Using Random Forest

Random Forest was selected as the classification method due to its ability to handle high-dimensional handcrafted features and its robustness against overfitting, which is especially important given the limited size of the Shenzhen CXR dataset. Unlike margin-based classifiers, Random Forest constructs multiple decision trees and aggregates their outputs, resulting in more stable and generalized predictions.

Key reasons for selecting Random Forest include:

- supports non-linear decision boundaries without complex tuning
- performs well on small-to-medium medical datasets
- resistant to noise and feature variability
- provides built-in feature importance for interpretability

This makes the model suitable for TB screening tasks where diagnostic transparency and reliability are required.

1) Hyperparameter Configuration

Instead of kernel and margin parameters used in SVM, Random Forest relies on tree-based settings. In this study, the model configuration was based on empirical testing in the notebook:

- `n_estimators = 200` (number of trees in the ensemble)
- `max_depth = None` (allow trees to expand fully)
- `criterion = "gini"` (split quality metric)
- `random_state = 42` (ensures reproducibility)

No grid search was applied due to dataset constraints; however, the chosen parameters provided stable performance across validation runs.

2) Training Strategy

Since Random Forest is a classical machine learning algorithm, the training process differs from iterative optimization used in neural networks.

The model was trained using:

- a single fitting step without epochs
- the full training set processed at once
- stratified 80–10–10 splitting to preserve class balance
- no gradient updates or learning rate adjustments

Final predictions were generated through **majority voting** across all trees, where the class receiving the highest number of votes was selected as the output label, and probability scores were derived from the proportion of tree votes.

IV. EXPERIMENTAL SETUP

A. Hardware and Software Configuration

All experiments were conducted on a standard computing environment without GPU acceleration to ensure reproducibility and low-resource feasibility. The system utilized a consumer-grade laptop running Windows, demonstrating that the Random Forest-based TB detection pipeline can operate efficiently without specialized hardware.

System specifications:

- **Processor:** Intel Core i5 CPU
- **Memory:** 8 GB RAM
- **GPU:** None (CPU-only execution)
- **Operating System:** Windows 10 or MacOS

Software environment:

- **Python 3.x**
- **Libraries used:**
 - OpenCV for preprocessing and segmentation
 - Scikit-image for GLCM texture extraction
 - Scikit-learn for Random Forest training and evaluation
 - NumPy and Pandas for numerical computation and data handling

This setup confirms that the method is suitable for deployment in low-resource clinical settings.

B. Dataset and Splitting Strategy

The Shenzhen TB Chest X-ray dataset was used as the source of radiographic images for classification. To ensure balanced evaluation, the dataset was divided using a stratified split, preserving the proportion of TB-positive and normal samples across all subsets.

Data allocation:

- **80%** for training
- **10%** for validation
- **10%** for testing

All images were resized, segmented, and converted into feature vectors prior to classification, ensuring consistency across the pipeline.

C. Evaluation Metrics

Model performance was assessed using multiple standard diagnostic metrics recommended for medical

imaging studies. These measures provide insight into both overall accuracy and the ability to correctly detect TB-positive cases.

The following metrics were computed:

- **Accuracy:** proportion of correctly classified samples
- **Precision:** likelihood that predicted positive cases are truly TB
- **Recall (Sensitivity):** ability to detect actual TB-positive images
- **Specificity:** ability to correctly identify non-TB cases
- **F1-score:** harmonic mean of precision and recall
- **ROC-AUC:** discriminatory capability across threshold variations

These metrics were generated using scikit-learn evaluation functions and reported on the held-out test set.

D. Hyperparameter Configuration

The Random Forest classifier was implemented using fixed parameters based on empirical testing rather than automated search, due to the limited dataset size and stable validation results.

Final configuration:

- `n_estimators = 200`
- `max_depth = None`
- `criterion = "gini"`
- `random_state = 42`

No pruning or regularization was required, and training completed in a single pass without iterative optimization, reflecting the efficiency of tree-based methods.

V. RESULT AND DISCUSSION

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A. Classification Performance

The Random Forest classifier achieved strong overall performance on the held-out test set, demonstrating its effectiveness in distinguishing between TB-positive and normal chest X-ray images based on handcrafted intensity, texture, and shape features. The model obtained an **overall accuracy of 0.93**, indicating that 93% of all test samples were correctly classified. Precision, recall, and F1-score values were consistently high for both classes, confirming the classifier's balanced behavior across normal and TB categories.

Key classification metrics:

- **Accuracy:** 0.93

- **Precision:**
 - Normal: 0.95
 - TB Positive: 0.90
- **Recall:**
 - Normal: 0.90
 - TB Positive: 0.95
- **F1-score:**
 - Normal: 0.92
 - TB Positive: 0.93
- **Macro F1:** 0.92
- **AUC:** 0.9702

The confusion matrix further illustrates strong classification performance:

	Predicted Normal	Predicted TB
Actual Normal	72	8
Actual TB	4	76

The model shows higher recall for TB cases (0.95), meaning it is effective at identifying positive cases—a crucial property for medical screening, where false negatives must be minimized. Precision for the Normal class (0.95) is also high, reflecting a low rate of false alarms.

B. ROC-AUC Analysis

The model achieved an **AUC score of 0.9702**, demonstrating excellent discriminative capability across decision thresholds. An AUC close to 1.0 indicates that the Random Forest classifier can reliably distinguish between TB-positive and normal images even before a threshold is applied. This is especially valuable in medical scenarios where sensitivity-specificity trade-offs must be adjusted depending on clinical priorities.

C. Comparison Between Cropping and Segmentation Approaches

Two preprocessing methodologies were tested during experimentation:

- (1) **crop-based ROI extraction**, and
- (2) **lung segmentation before feature extraction**.

Although segmentation intuitively seems advantageous because it isolates the anatomical lung region, empirical results showed that **cropping produced more consistent and higher classification performance**, particularly in precision, recall, and F1-score.

Observed differences between methods:

- **Cropping approach:**

- Higher overall performance across precision, recall, and F1
 - Produced the best balanced accuracy (0.93)
 - More stable feature representations due to fewer segmentation artifacts
- **Segmentation approach:**
 - Achieved a slightly higher AUC in some trials
 - But overall precision/recall/F1 were lower and less stable
 - Inaccurate masks occasionally reduced texture and shape quality

This aligns with the team's internal reflection, where both members observed that segmentation-based preprocessing resulted in small performance declines compared to simple cropping. While segmentation is theoretically cleaner, the crop-based method empirically outperformed due to producing more consistent features across images.

D. Discussion and Interpretation

The results indicate that **Random Forest combined with handcrafted features remains a strong and computationally efficient approach for TB detection**, capable of achieving over 93% accuracy without deep learning or GPU resources. The performance metrics demonstrate robustness, especially in detecting TB-positive samples, which is critical in real-world clinical screening scenarios.

The slight performance drop from segmentation-based preprocessing suggests that segmentation quality strongly influences downstream feature extraction. Imperfect masks may remove diagnostically relevant structures or introduce noise, leading to reduced classification reliability. Conversely, the crop-based method preserved more global radiographic information, enabling Random Forest to exploit both lung and contextual features.

These findings reinforce the importance of preprocessing strategy selection and demonstrate that, in some cases, **simpler methods outperform more complex ones** when dataset size is limited.

VI. CONCLUSION

This study successfully developed a complete tuberculosis detection pipeline based on handcrafted feature extraction and Random Forest classification, demonstrating that classical machine learning methods can still achieve high accuracy and reliability in medical imaging tasks. The pipeline delivered strong diagnostic performance, reaching

93% accuracy, 0.93 F1-score, and an **AUC of 0.9702**, confirming that the model is able to learn discriminative radiological patterns effectively from chest X-ray images. The high recall for TB-positive cases (0.95) further highlights the system's sensitivity-first performance—an essential quality for screening applications in which missed detections can have serious clinical consequences.

A key strength of this approach is its lightweight, resource-efficient implementation. The entire workflow—ranging from preprocessing and feature extraction to model training—operated smoothly on a standard CPU laptop without any GPU acceleration. This demonstrates the practicality of the system for deployment in low-resource or rural healthcare environments where advanced hardware is often unavailable. Unlike deep learning models that require large datasets and intensive computation, the proposed Random Forest pipeline remains interpretable, fast, and easy to integrate into existing diagnostic procedures.

Beyond its immediate performance, the model also shows promising generalization capability. By relying on statistical, texture, and shape-based features rather than high-capacity neural architectures, the system is less prone to overfitting, making it suitable for small or moderately sized medical datasets. The comparative experiments further revealed that preprocessing strategies such as cropping versus segmentation influence performance, providing insights that can guide future improvements.

Overall, this work demonstrates that a classical, well-engineered machine learning pipeline can meaningfully support TB screening workflows by offering fast, sensitive, and cost-effective preliminary detection. While not intended to replace radiologists, the system has the potential to act as an assistive decision-support tool—prioritizing suspected TB cases, reducing screening workload, and improving early detection in communities where expert interpretation is limited. Future research can expand this framework with larger datasets, more advanced segmentation techniques, or hybrid methods that combine handcrafted features with deep representations to further enhance robustness and clinical applicability.

VII. FUTURE WORK

Future improvements can further enhance the performance, scalability, and clinical readiness of this tuberculosis detection system. Although the current Random Forest-based approach provides strong accuracy and low hardware requirements, several extensions can significantly advance the pipeline.

A. Deep Learning Integration

A major direction for future work is the adoption of **Convolutional Neural Networks (CNNs)** to enable end-to-end feature learning directly from raw chest X-ray images. CNNs can automatically extract hierarchical radiological patterns, potentially improving sensitivity and robustness on more diverse datasets.

B. Multi-Dataset Training

To enhance generalization, the system can be expanded to incorporate **multiple TB datasets**, such as Montgomery, NIH, or private clinical collections. Training across multi-institutional data would reduce dataset bias and strengthen model performance on unseen populations.

C. Real-Time Inference System

Implementing a **real-time screening module** would enable near-instantaneous predictions during clinical workflows. This includes optimizing preprocessing pipelines, batching inference, and developing an API-based deployment layer for hospital systems.

D. Explainability and Clinical Transparency

To increase trust and interpretability, post-hoc explainability methods can be integrated, such as:

- **Grad-CAM** for heatmap-based region attribution
- **LIME** for local feature explanation
- **SHAP** for global feature importance

These tools would allow clinicians to visualize why the model predicted TB, making the system safer for real-world use.

E. Mobile and Low-Cost Deployment

Finally, deploying the model on **mobile or edge devices** could significantly expand access to screening in remote or low-resource regions. Techniques such as model quantization, lightweight inference engines, and on-device image preprocessing can enable portable, offline screening compatible with field conditions.

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Their combined contributions—including preprocessing design, model experimentation, analysis, and report preparation—were essential to the successful completion of this study.

REFERENCES

- [1] World Health Organization, “Global Tuberculosis Report 2023.” Available: <https://www.who.int/teams/global-tuberculosis-programme/tb-reports>
- [2] C.-Y. Hong et al., “Time-Trend Analysis of Tuberculosis Diagnosis in Shenzhen, China between 2011 and 2020,” *Frontiers in Public Health*, vol. 11, 2023
- [3] T. Rahman et al., “Reliable Tuberculosis Detection Using Chest X-Ray with Machine Learning, Segmentation and Visualization,” *IEEE Access*, vol. 8, pp. 191586–191601, 2020.
- [4] K. K. Goswami et al., “Machine Learning Classification of Tuberculosis Chest X-Rays,” *Cureus*, 2023.
- [5] K. K. Goswami et al., “Machine Learning Classification of Tuberculosis Chest X-Rays,” *Cureus*, 2023.
- [6] C.-F. Chen et al., “A Machine Learning-Based Algorithm for Pulmonary Tuberculosis Detection,” *Scientific Reports*, 2024.
- [7] Goswami, K. K., Paul, S., & Sengupta, R. (2023). Machine learning classification of tuberculosis chest X-rays. *Cureus*. <https://www.cureus.com/articles/115659-machine-learning-classification-of-tuberculosis-chest-x-rays>
- [8] Qin, Z., Chen, S., & Liu, X. (2022). Performance of chest radiography for tuberculosis screening in community-based settings. *Scientific Reports*. <https://www.nature.com/articles/s41598-022-12898-5>
- [9] World Health Organization. (2023). Global Tuberculosis Report 2023.
- [10] Goswami, K. K., Paul, S., & Sengupta, R. (2023). Machine learning classification of tuberculosis chest X-rays. *Cureus*. <https://www.cureus.com/articles/115659-machine-learning-classification-of-tuberculosis-chest-x-rays>
- [11] Lakhani, P., & Sundaram, B. (2017). Deep learning at chest radiography: Automated classification of pulmonary tuberculosis by using convolutional neural networks. *Radiology*.
- [12] Qin, Z., Chen, S., & Liu, X. (2022). Performance of chest radiography for tuberculosis screening in community-based settings. *Scientific Reports*. <https://www.nature.com/articles/s41598-022-12898-5>
- [13] Rahman, T., et al. (2020). Reliable tuberculosis detection using chest X-ray with machine learning, segmentation, and visualization. *IEEE Access*. <https://ieeexplore.ieee.org/document/9187161>
- [14] World Health Organization. (2023). Global Tuberculosis Report 2023. <https://www.who.int/teams/global-tuberculosis-programme/tb-reports> Lakhani, P., & Sundaram, B. (2017). Deep learning at chest radiography: Automated classification of pulmonary tuberculosis. *Radiology*.
- [15] Qin, Z., Chen, S., & Liu, X. (2022). Performance of chest radiography for tuberculosis screening in community settings. *Scientific Reports*. <https://www.nature.com/articles/s41598-022-12898-5>
- [16] World Health Organization. (2023). Global Tuberculosis Report 2023. <https://www.who.int/teams/global-tuberculosis-programme/tb-rep>